

Question	Marking Guidance	Mark	Comments
03.1	Second box ticked; Answer key: B – statements 1, 2 and 4	1 (AO1)	

Question	Marking Guidance	Mark	Comments
03.2	A correct answer of $9(.144) \times 10^8 = 2 \text{ marks;;}$ Answer not in standard form (914 400 000) = 1 mark OR Answer in incorrect standard form = 1 mark OR 1.016×10^8 (everything else correct except failed to multiply by 9) = 1 mark OR 2.286×10^8 (everything else correct but failed to take the plating of 0.25 cm^3 into account) = 1 mark OR 9.432×10^8 (discounted 238 as anomalous, all other calculations correct) = 1 mark	2 (2 x AO2)	

Question	Marking Guidance	Mark	Comments
03.3	30;	1 (AO2)	Accept correct number of generations from incorrect response to 03.2, Table 1 on page 11 will help with this Examples of how to use logs to solve $2^x = 9.14 \times 10^8$ $\ln(2^x) = \ln(9.14 \times 10^8)$ $x \cdot \ln(2) = \ln(9.14 \times 10^8)$ $x = \frac{\ln(9.14 \times 10^8)}{\ln(2)}$ $x = 30$ OR $2^x = 9.144 \times 10^8$ $\log_2(9.144 \times 10^8) = 29.768 \text{ (so 30 generations)}$

Table 1 for question 03.3

Number of generations	Standard form	Population size
1	2×10^0	2
2	4×10^0	4
3	8×10^0	8
4	1.6×10^1	16
5	3.2×10^1	32
6	6.4×10^1	64
7	1.28×10^2	126
8	2.56×10^2	256
9	5.12×10^2	512
10	1.024×10^3	1024
11	2.048×10^3	2048
12	4.096×10^3	4096
13	8.192×10^3	8192
14	1.638×10^4	16384
15	3.277×10^4	32768
16	6.553×10^4	65536
17	1.311×10^5	131072
18	2.621×10^5	262144
19	5.243×10^5	524288
20	1.049×10^6	1048576
21	2.097×10^6	2097152
22	4.194×10^6	4194304
23	8.388×10^6	8388608

Number of generations	Standard form	Population size
24	1.678×10^7	16777216
25	3.355×10^7	33554432
26	6.711×10^7	67108864
27	1.342×10^8	134217728
28	2.684×10^8	268435456
29	5.369×10^8	536870912
30	1.073×10^9	1073741824
31	2.147×10^9	2147483648
32	4.295×10^9	4294967296
33	8.590×10^9	8589934592
34	1.718×10^{10}	17179869184
35	3.436×10^{10}	34359738368
36	6.872×10^{10}	68719476736
37	1.374×10^{11}	
38	2.749×10^{11}	
39	5.498×10^{11}	
40	1.100×10^{12}	
41	2.199×10^{12}	
42	4.398×10^{12}	
43	8.796×10^{12}	
44	1.759×10^{13}	
45	3.518×10^{13}	

Question	Marking Guidance	Mark	Comments
03.4	1. The concentration (of chloramphenicol) was too low/dilute (to kill all) OR There were too many <i>M. luteus</i>/bacteria for chloramphenicol/antibiotic to kill; 2. (Contamination with) other/resistant bacteria; 3. Mutation (occurred);	2 max (2 x AO3)	Accept as an additional mark point 4. Chloramphenicol/antibiotic not (evenly) mixed in the liquid culture/bottle
